(b)
Python source code：

```
import gurobipy as gp
from gurobipy import GRB

# 创建求解 primal 问题的模型
model = gp.Model("WorkerScheduling")

# 决策变量
x = model.addVars(5, lb=0, vtype=GRB.CONTINUOUS, name="x")
y = model.addVars(5, lb=0, vtype=GRB.CONTINUOUS, name="y")

# 约束条件
model.addConstr(x[0] == 50)
for t in range(2, 6):
    model.addConstr(x[t - 1] == 0.95 * x[t - 2] + y[t - 2])

model.addConstr(160 * x[0] - 50 * y[0] >= 6000)
model.addConstr(160 * x[1] - 50 * y[1] >= 7000)
model.addConstr(160 * x[2] - 50 * y[2] >= 8000)
model.addConstr(160 * x[3] - 50 * y[3] >= 9500)
model.addConstr(160 * x[4] - 50 * y[4] >= 11000)

# 目标函数
obj = gp.LinExpr()
for i in range(5):
    obj += 2000 * (x[i])
model.setObjective(obj, GRB.MINIMIZE)

# 求解模型
model.optimize()

# 输出结果
if model.status == GRB.OPTIMAL:
    print("primal 问题最优解:")
    for i in range(5):
        print(f"month{i}:fulltime_workers = {x[i].x},
month{i}:learning_workers = {y[i].x}")
    print(f"目标函数值: {model.objVal}")

else:
```

```python
    print("没有找到 primal 问题的最优解.")

print("对偶问题最优解: ")
print(model.getAttr("Pi",model.getConstrs()))
duel_solution=model.getAttr("Pi",model.getConstrs())
duel_ObjVal=50*duel_solution[0]+6000*duel_solution[5]+7000*duel_solution[6]+8000*duel_solution[7]+9500*duel_solution[8]+11000*duel_solution[9]
print("对偶问题的目标函数值:")
print(duel_ObjVal)

if duel_ObjVal==model.ObjVal:
    print("强对偶性成立")
else:
    print("强对偶性不成立")
```

the output file:

```
PS D:\python学习\线性规划> & d:/python学习/线性规划/.venv/Scripts/python.exe d:/python学习/线性规划/1.py
Set parameter Username
Academic license - for non-commercial use only - expires 2025-09-24
Gurobi Optimizer version 11.0.3 build v11.0.3rc0 (win64 - Windows 11.0 (22621.2))

CPU model: 13th Gen Intel(R) Core(TM) i7-1360P, instruction set [SSE2|AVX|AVX2]
Thread count: 12 physical cores, 16 logical processors, using up to 16 threads

Optimize a model with 10 rows, 10 columns and 23 nonzeros
Model fingerprint: 0xb17731e3
Coefficient statistics:
  Matrix range     [9e-01, 2e+02]
  Objective range  [2e+03, 2e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [5e+01, 1e+04]
Presolve removed 6 rows and 5 columns
Presolve time: 0.00s
Presolved: 4 rows, 5 columns, 12 nonzeros

Iteration    Objective       Primal Inf.    Dual Inf.      Time
       0   -2.6315789e+32   9.847862e+29   2.631579e+02      0s
       2    5.6435513e+05   0.000000e+00   0.000000e+00      0s

Solved in 2 iterations and 0.01 seconds (0.00 work units)
Optimal objective  5.643551314e+05
primal问题最优解:
month0:fulltime_workers = 50.0, month0:learning_workers = 0.0
month1:fulltime_workers = 47.5, month1:learning_workers = 8.453168094062995
month2:fulltime_workers = 53.578168094062995, month2:learning_workers = 11.450137901001604
month3:fulltime_workers = 62.34939759036145, month3:learning_workers = 9.518072289156628
month4:fulltime_workers = 68.75, month4:learning_workers = 0.0
目标函数值: 564355.131368849
对偶问题最优解:
[3900.0, 2000.0, 0.0, -481.9277108433735, -598.0548700827406, 0.0, 0.0, 9.63855421686747, 11.961097401654813, 16.23784293801713]
对偶问题的目标函数值:
564355.131368849
强对偶性成立
PS D:\python学习\线性规划>
```

(d)

Following the above code, the problem handling code is as follows:

```
# 更换 april 和 may 的对应 b 参数
constr_1=model.getConstrByName("April")
constr_2=model.getConstrByName("May")
model.setAttr(GRB.Attr.RHS,[constr_1],[9000])
model.setAttr(GRB.Attr.RHS,[constr_2],[10000])

model.optimize()

# 输出结果
if model.status == GRB.OPTIMAL:
    print("修改后的问题最优解:")
    for i in range(5):
        print(f"month{i}:fulltime_workers = {x[i].x},
month{i}:learning_workers = {y[i].x}")
    print(f"目标函数值: {model.objVal}")

else:
    print("没有找到修改后问题的最优解.")
```

the output file:

```
Solved in 0 iterations and 0.00 seconds (0.00 work units)
Optimal objective  5.421367397e+05
修改后的问题最优解:
month0:fulltime_workers = 50.0, month0:learning_workers = 0.0
month1:fulltime_workers = 47.5, month1:learning_workers = 7.509634925243148
month2:fulltime_workers = 52.63463492524315, month2:learning_workers = 8.430831760778055
month3:fulltime_workers = 58.433734939759034, month3:learning_workers = 6.987951807228917
month4:fulltime_workers = 62.5, month4:learning_workers = 0.0
目标函数值: 542136.7397300045
```